

Course Handout

Institute/School Name	Chitkara University Institute of Engineering and Technology		
Department Name	Department of Computer Science & Engineering		
Programme Name	Bachelor of Engineering (B.E.), Computer Science & Engineering		
Course Name	Problem Solving using Python Programming	Session	July 2025- Dec-2026
Course Code	24CSE0101	Semester/Batch	1 st /2025
L-T-P(Per Week)	2-0-6	Course Credits	05
Pre-requisite	NA	NHEQF Level	4.5
Course Coordinator	Dr. Vibhu Sharma	SDG Number	4, 9

1. Objectives of the Course

Python is an open-source, high-level, dynamically-typed, portable, expressive, easy to learn, and well known programming language. Python is available as an open-source offering and is associated with large global community support. Top companies such as Google, Yahoo!, Pinterest, Disney, Nokia, IBM have python as one of their main programming language. This course provides a wide scope of learning & understanding of python programming . The main objectives of the course are:

1. To understand the basics of Python (operations, control structures, data types, etc.)
2. Apply various data structures (list, tuple, string, dict.) for problem solving, fostering Industry, and Innovation.
3. Design and implement logic-building programs using Python constructs to enhance Quality Education.
4. Understand the object-oriented program design and development.
5. Understand class inheritance and polymorphism for advanced programming skills, aligned with Quality Education.
6. Design and develop real-time applications using Python GUI programming, promoting Industry, Innovation, and Infrastructure.

2. Course Learning Outcomes (CLOs)

Student should be able to:

CLOs	Course Learning Outcome	Program Outcomes (PO)	NHEQF Level Descriptor	No. of Lectures
CLO01	Understand Python lexical features, structures, and flow control for logic building.	PO1, PO3, PO9, PO10	Q1, Q2	14
CLO02	Apply decision statements and loops in Python to solve complex problems.	PO1, PO2, PO3, PO4	Q3, Q6	14
CLO03	Implement Python functions to facilitate code reuse and manipulate strings.	PO3, PO5, PO11	Q3	14
CLO04	Create and manipulate structured data using lists, sets, tuples, and dictionaries to develop real-world solutions.	PO2, PO3, PO5, PO9, PO11	Q1, Q2	14
CLO05	Demonstrate the use of built-in functions to navigate the file system and manage data efficiently.	PO3, PO4, PO5, PO7, PO9, PO10, PO11	Q3	16
CLO06 (Only for lab components)	Develop real-world applications using the Object-Oriented Approach and Python GUI programming.	PO3, PO5, PO7, PO9, PO10, PO11	Q3, Q6	16
Total Contact Hours				88

CLO-PO Mapping

CLO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	Type of Assessment's
CLO01	H		H						M	H		Formative, Summative
CLO02	H	H	H	M								Formative, Summative
CLO03			H								M	Formative, Summative
CLO04		H	H		M				H		M	Formative, Summative
CLO05			M	H	H		M		H	M	L	Formative, Summative
CLO06		H	H		M		M		H	M	H	Formative

H=High, M=Medium, L=Low

3. Recommended Books:

- B01:** Martin C. Brown, "The Complete Reference", MacGraw Hill Education 2021.
B02: Dr. R. Nageswara Rao, "Core Python Programming", Dreamtech Press 2019.
B03: Thareja, Reema, "Python Programming: Using Problem Solving Approach". Oxford University Press c2017.
B04: Ravichandran, R. Saravanan, "Learn to Master Python", Staredu Solution, c2018.
B05: Paul Barry, "Head First Python"
B06: Roberto Tamassia, Michael H Goldwasser, Michael T Goodrich, "Data Structures and Algorithms in Python", 1st Edition, Wiley India Pvt Ltd, 2016. ISBN-13: 978-8126562176

4. Other readings and relevant websites:

SerialNo	Link of Journals, Magazines, websites and Research Papers
1.	https://www.python.org/
2.	https://learnpython.org/
3.	https://pythoninstitute.org/

5. Recommended Tools and Platforms

Python 3 IDLE, Anaconda 3 (Jupyter notebook 3.6.0), Google colab, Replit

6. Course Plan: Theory+ Lab Plan

Theory Plan

Lect. No.	Topic(s)
1-3	Introduction to Python programming: Python environment setup, identifiers, reserved words, lines and indentation, multi-line statements, comments, quotation, input/output, output formatting (Integers and string). Fundamentals: Variables (simple, assigning multiple values, output variables, global/local variables)
4-7	Python decision making: If else and Nested, If else and elif. Match statement, Introduction to while and for loop.
8-11	Concept of nested looping. For loop and range method, Break, Continue and Pass, Pattern Design.
12-14	Defining and calling Functions, Type of arguments (required, keyword, default, variable length), Scope Rules (Global/local Statements), Passing Arguments (call by value/call by reference), Recursive Functions, Lambda Expressions.
15-17	Lists: Introduction to Lists, Operations on Lists, Storage Structure, Re-sizing and Looping on Lists, Indexing and Slicing, Passing variables through Functions, passing Lists through Functions.
ST1 (Syllabus=30% (Lect. No. 1-17))	
18-21	List Comprehension, map, filter & reduce functions, Two Dimensional Lists: Introduction and Storage, Jagged Lists.
22-24	String- Introduction, methods, comprehension and its relative properties.
25-27	Tuples, Set and Dictionaries- Introduction, methods, comprehension and its relative properties.
ST2 (Syllabus=60% (Lect No. 1-27))	
28-32	Modules: Creating Modules, Variables in Modules, Imports and Attributes, Namespaces, Reloading, Built in modules, Generating Random values.
32-36	Files and Directories, File I/O, File positioning, File operators.
37-40	Exception Handling: Introduction to Exception, Exception Handling, Except clause, Try, finally clause, User Defined Exceptions.
41-44	Exploring Object-Oriented Programming In Python-Introduction, Creating classes, objects, attributes. Constructors., Idea of inheritance, Superclass And SubClass. Overriding methods, Object class, Isinstance function.
End Term Exam (Syllabus=100% (Lect No. 1-44))	

Lab Plan

Lab No.	Topic(s)	Problem Statement
1-2	Anaconda Installation, Running Python Programs, Writing Python Scripts with Jupyter Notebook, and also make familiar with various platforms used for python programming, Implementation of basic concepts of python	Set up your environment (Anaconda or Google Colab). Create a new Jupyter Notebook named 'hello_tools.ipynb'. Print 'Hello, Python!', check Python version using sys.version, and create a simple addition program that takes two inputs from the user and prints the sum. This ensures the environment is set up and basic execution is understood.
3-4	Data types (text type, numeric, sequence, mapping, set), Python basic operators- arithmetic, comparison, assignment operators, logical operators, mix type conversion, integer to float and float to integer conversion.	Create a notebook to define variables of different data types (str, int, float, bool, list, tuple, set, dict). Print each variable and its type, perform type conversions between int, float, and string, and demonstrate mixed-type expressions. Show arithmetic, comparison, logical, and assignment operators with examples.
5-6	Coding problems related to conditional statements (if, elif, nested if else) and match statement.	Write a program that calculates travel fare based on age and day of the week. Use if/elif to assign fares for age groups and match statement to apply weekend discounts. Print the final fare from a base price, ensuring conditionals and match statements are applied correctly.
7-9	Coding problems related to for/while loops, For loop and range method, Break, continue and pass. Nested loops and pattern designs	Develop a program to print multiplication tables using for loops, sum even numbers with while loops, and demonstrate break and continue. Create nested loop patterns like triangles, pyramids, and hollow squares for hands-on loop practice.
10-13	Implementation related to Functions: User Defined and Built in functions(Python mathematical functions, random number functions), Passing Arguments(call by value/call by reference), Recursive Functions, Lambda Expressions.	Create a set of functions: circle_area using math.pi, compound_amount with default parameters, and roll_dice using random.randint. Implement recursive factorial and Fibonacci functions, and demonstrate lambda with map and filter to manipulate lists.
14-16	Implementation of lists. Creating empty list, Initializing list, list indexing and slicing operations, input method, Concatenating multiple lists, generating range based lists. list based methods like append(), extend(), insert(), index(), count(), sort(), reverse(), pop() and nested lists.	Build a grocery list manager that starts empty and lets the user add items until 'done'. Insert priority items, extend with predefined lists, use index and count, sort, reverse, and pop items. Create a nested list of categories and print them neatly.
17-19	Implementation of 2-D list with basic operations.	Create a 2-D list for student marks, compute row-wise totals and averages, column-wise averages, and generate the transpose of the list. Demonstrate indexing and traversal of 2-D structures.
20-23	Python strings- accessing values in strings, updating strings, string special characters, string formatting operators, triple quotes, built in string methods.	Develop a text analyzer that takes a sentence and applies string methods like capitalize, lower, upper, strip, replace, find, index, count, startswith, endswith, split, isalpha, isdigit, isalnum, islower, and isupper. Identify the longest and shortest words, and print min and max characters.
24-26	Implementation of Sets and relative methods; Implementation of Tuples and relative methods; Creating empty dictionaries, initializing dictionaries, Accessing dictionary items, merging, and deletion, dictionary methods, Python date and time functions.	Create an attendance tracker using sets to store unique attendees, tuples for fixed seat coordinates, and dictionaries mapping roll numbers to names. Demonstrate set operations, tuple immutability, and dictionary methods like keys, values, items, copy, update, and pop. Add date/time stamping for sign-ins.
27-29	Python modules-import, from... import statements, dir function, Generating Random values (random module), File handling- Writing to a file, Reading from a file, Resetting the file current position, fseek.	Build a quiz game that stores MCQs in a list of dictionaries. Use random to pick questions, write results to a file with timestamp, read back results, and demonstrate file cursor movement using seek.
30-32	Exploring Object-Oriented Programming In Python: Creating classes, objects, attributes, Constructors, Inheritance, Overriding methods, Operator Overloading.	Create classes for contacts with inheritance and method overriding. Implement a contact book with methods to add, find, and merge contacts using operator overloading. Demonstrate isinstance checks and make the class iterable.
33-34	Introduction to GUI Programming (Tkinter), Creating root windows, Adding Widgets, Layout Management, Event Handling.	Build a Tkinter app with labels, entry boxes, buttons, and status labels. Use grid layout, bind events to actions like adding and searching contacts, and update the GUI dynamically.
35-36	Working with Tkinter Widgets: Labels, Buttons, Entry Widgets, Text Widgets, Frames, Menus.	Enhance the Tkinter app to include menus, multi-line text entry, frames to group widgets, and event handling for menu selections. Display outputs in label or text widgets.
37-40	Connecting Python with Oracle 11G (or SQLite) - Performing insertion, updating, and deletion operations using Python GUI.	Integrate the contact book GUI with a database. Implement buttons to insert, update, delete, and display records. If Oracle 11g is unavailable, use SQLite for demonstration.
41-44	Final Project - Building a complete Python GUI Application.	Combine OOP, Tkinter GUI, and database connectivity to build a complete functional contact management system. Ensure the app can add, update, delete, search, and display contacts with a user-friendly interface.

7. Delivery/Instructional Resources
Theory Plan:

Lect. No.	Topics	CLO	Book No, CH No, Page No	TLM	ALM	Web References	Audio-Video
1-3	Introduction to Python programming. Python environment setup, identifiers, reserved words ,lines and indentation, multiline statements, comments, quotation, input/output, output formatting(Integers and string). Fundamentals: Variables(simple, assigning multiple values, output variables, global/local variables)	CLO01	B01, CH 1, Page no 1-17 B02, CH 2, Page No 19-24 B03, CH 03, Page No 83-124	Lecture	Quiz, Test questions	https://www.anaconda.com/ https://www.learnpython.org/ https://docs.python.org/3/tutorial/introduction.html	https://nptel.ac.in/courses/106106182
4-7	Python decision making- If else and Nested, If else and elif. Match statement, Introduction to while and for loop.	CLO02	B02, CH 6, Page No 117-149 B03, CH 04, Page No 137-179	Lecture	Quiz, Puzzles, Tests	https://docs.python.org/3/tutorial/controlflow.html	https://nptel.ac.in/courses/106106182
8-11	Concept of nested looping. For loop and range method, Break, Continue and Pass, Pattern Design.	CLO02	B02, CH 6, Page No 117-149	Lecture	Quiz, Brain Storming Sessions	https://docs.python.org/3/tutorial/controlflow.html https://docs.python.org/3/tutorial/controlflow.html#for-statements	https://nptel.ac.in/courses/106106182 https://nptel.ac.in/courses/106106182
12-14	Defining and calling Functions, Type of arguments (required, keyword, default, variable length), Scope Rules(Global/local Statements), Passing Arguments(call by value/call by reference), Recursive Functions, Lambda Expressions.	CLO03	B01, CH 4, Page No 81-98 B02, CH 9, Page No 237-280	Lecture	Quiz, Peer Review	https://docs.python.org/3/tutorial/controlflow.html#more-on-defining-functions	https://www.coursera.org/lecture/python-genomics/lecture-5-1-functions-part-1-5-54-PGXxZ https://www.coursera.org/lecture/python-genomics/lecture-5-2-functions-part-2-8-20-yIbSK
15-17	Lists: Introduction to Lists, Operations on Lists, Storage Structure, Resizing and Looping on Lists, Indexing and Slicing, Passing variables through Functions, passing Lists through Functions.	CLO04	B02, CH 10, Page No 283-306	Lecture	Test Questions, Quiz	https://docs.python.org/3/tutorial/datastructures.html	https://nptel.ac.in/courses/106106182
18-21	List Comprehension, map, filter & reduce functions, Two Dimensional Lists: Introduction and Storage, Jagged Lists.	CLO04	B02, CH 10, Page No 283-306 B04, CH 08, Page No 238	Lecture	Test Questions	https://docs.python.org/3/tutorial/datastructures.html#list-comprehensions	https://nptel.ac.in/courses/106106182
22-24	String- Introduction, methods, comprehension and its relative properties.	CLO04	B02, CH 8, Page No 207-235	Lecture	Test Questions	https://docs.python.org/3/tutorial/controlflow.html#documentation-strings	https://www.coursera.org/lecture/python-representation/python-strings-bbBTs https://www.coursera.org/lecture/python-representation/python-strings-bbBTs

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25-27	Tuples, Set and Dictionaries- Introduction, methods, comprehension and its relative properties.	CLO04	B02, CH 10, Page No 307-318 B04, CH 09, Page No 257-281	Lecture	Test Questions	https://docs.python.org/3/tutorial/datastructures.html https://docs.python.org/3/tutorial/datastructures.html#tuples-and-sequences	https://nptel.ac.in/courses/106106182
28-32	Modules: Creating Modules, Variables in Modules, Imports and Attributes, Namespaces, Reloading, Built in modules, Generating Random values.	CLO05	B01, CH 5, Page No 101-111	Lecture	Brain Storming Sessions, Focused Listing	https://docs.python.org/3/tutorial/modules.html	https://www.coursera.org/lecture/python-genomics/lecture-6-modules-and-packages-10-32-rejwk
32-36	Files and Directories, File I/O, File positioning, File operators.	CLO05	B03, CH 7, Page No 289-307	Lecture	Brain Storming Sessions	https://python-course.eu/python-tutorial/file-management.php https://docs.python.org/3/tutorial/inputoutput.html#reading-and-writing-files	https://www.coursera.org/lecture/data-analytics-accountancy-1/python-file-io-csSim https://www.coursera.org/lecture/accounting-data-analytics-python/5-1-python-file-io-R6L3h
37-40	Exception Handling: Introduction to Exception, Exception Handling, Except clause, Try, finally clause, User Defined Exceptions.	CLO05	B03, CH 12, Page No 480-494	Lecture	Leading Question	https://docs.python.org/3/tutorial/errors.html https://ehmatthes.github.io/pcc/cheatsheets/README.html#chapter-10-files-and-exceptions	https://www.coursera.org/projects/exception-handling-in-python
41-44	Exploring Object-Oriented Programming In Python-Introduction, Creating classes, objects, attributes. Constructors., Idea of inheritance, Superclass And SubClass. Overriding methods, Object class, Isinstance function.	CLO06	B03, CH 09, Page No 400-426 B03, CH 10, Page No 436-447	Lecture	Leading Question, Quiz, One Minute Paper	https://docs.python.org/3/tutorial/classes.html#highlight=object%20oriented	https://www.coursera.org/learn/object-oriented-python

Lab Plan:

Lab No.	Experiment	CLO	TLM	ALM	Web References	Audio-Video
1-2	Set up environment (Anaconda/Colab), create and run Python scripts in Jupyter Notebook. Print messages, check Python version, and perform basic arithmetic input/output.	CLO01	Demonstration method using a simulation or a tool	Quiz, Group Discussion	https://learning.anaconda.com/bundles/introduction-to-python-programming ; https://www.anaconda.com	https://nptel.ac.in/courses/106106182
3-4	Create variables for all Python data types, print their type, perform conversions, and demonstrate arithmetic, comparison, logical, and assignment operators.	CLO01	Discussion	Quiz, Student Created PPT, Flowcharts	https://www.learnpython.org/ ; https://www.geeksforgeeks.org/python-programming-language/	https://nptel.ac.in/courses/106106182

5-6	Write program to calculate travel fare using if/elif and match-case statements with conditions for age and weekend discounts.	CLO02	Demonstration method using a simulation or a tool	Quiz, Test Questions	https://www.learnpython.org/https://docs.python.org/3/tutorial/index.html#control-flow	https://nptel.ac.in/courses/106106182
7-9	Develop programs with for and while loops to print tables, sum even numbers, and use break/continue. Create patterns with nested loops.	CLO02	Lecture	Test Questions	https://docs.python.org/3/tutorial/index.html#control-flow#for-statements	https://nptel.ac.in/courses/106106182
10-13	Create functions using math and random modules, implement recursion, and demonstrate lambda with map and filter.	CLO03	Lecture	Test Questions	https://docs.python.org/3/tutorial/index.html#control-flow#more-on-defining-functions	https://www.coursera.org/lecture/python-genomics/lecture-5-1-functions-part-1-5-54-PGXxZ ; https://www.coursera.org/lecture/python-genomics/lecture-5-2-functions-part-2-8-20-yIbSK
14-16	Build grocery list manager demonstrating list creation, indexing, slicing, append, extend, insert, index, count, sort, reverse, pop, and nested lists.	CLO04	Lecture	Test Questions	https://docs.python.org/3/tutorial/datastructures.html	https://nptel.ac.in/courses/106106182
17-19	Create a 2-D list for student marks, compute totals, averages, and transpose the list.	CLO04	Lecture	Brain Storming Sessions	https://docs.python.org/3/tutorial/datastructures.html	https://nptel.ac.in/courses/106106182
20-23	Develop text analyzer applying string methods like capitalize, lower, upper, strip, replace, find, index, count, startswith, endswith, split, etc.	CLO04	Lecture	Brain Storming Sessions	https://docs.python.org/3/tutorial/index.html#control-flow#documentation-strings	https://www.coursera.org/lecture/python-representation/python-strings-bbBTs
24-26	Create attendance tracker using sets, tuples, and dictionaries with set operations, tuple immutability, dictionary methods, and date/time functions.	CLO04	Lecture	Leading Question, Group Discussion	https://docs.python.org/3/tutorial/datastructures.html#tuple ; https://docs.python.org/3/tutorial/datastructures.html#sets ; https://docs.python.org/3/tutorial/datastructures.html#dictionaries	https://nptel.ac.in/courses/106106182
27-29	Build quiz game using random module, file write/read operations, and file seek for cursor movement.	CLO03	Lecture	Brain Storming Sessions	https://docs.python.org/3/tutorial/modules.html ; https://python-course.eu/python-tutorial/file-management.php ; https://docs.python.org/3/tutorial/inputoutput.html#reading-and-writing-files	https://www.coursera.org/lecture/data-analytics-accountancy-1/python-file-i-o-cSSim
30-32	Create classes and inheritance with method overriding, operator overloading, and making classes iterable.	CLO06	Lecture and Inductive methods	Brain Storming Sessions	https://docs.python.org/3/tutorial/classes.html	https://www.coursera.org/learn/object-oriented-python
33-34	Build Tkinter GUI app with labels, entry widgets, buttons, and event handling.	CLO06	Lecture and Inductive methods	Brain Storming Sessions	https://docs.python.org/3/library/tkinter.html	https://www.coursera.org/projects/build-a-python-gui-with-tkinter
35-36	Enhance Tkinter app with menus, frames, multi-line text input, and additional event handling.	CLO06	Lecture and Inductive methods	Brain Storming Sessions	https://docs.python.org/3/library/tkinter.html	https://www.coursera.org/projects/gui-programming-login-system-python-tkinter
37-40	Integrate Tkinter app with database (Oracle or SQLite) for insert, update, delete, and select operations.	CLO06	Lecture and Inductive methods	Brain Storming Sessions	https://docs.python.org/3/library/tkinter.html	https://www.youtube.com/watch?v=YXPyB4XeYLA
41-44	Final project combining OOP, Tkinter GUI, and database connectivity for contact management.	CLO06	Lecture and Inductive methods	Brain Storming Sessions	https://docs.python.org/3/library/tkinter.html	https://www.youtube.com/watch?v=YXPyB4XeYLA

8. Remedial Classes

After every Sessional Test, different types of learners will be identified and special discussions will be planned and scheduled accordingly.

Action Plan for different types of learners:

Learner Type-I	Learner Type- II	Learner Type- III
Remedial Classes, Doubt Sessions, Guided Tutorials	Workshop, Doubt Session	Coding Competitions, Project

9. Self-Learning

Assignments to promote self-learning, survey of contents from multiple sources.

S.No	Topics	CLO	ALM	References/MOOCs
1	Installation, Conditional Statements, Functions, Loops and Iterations	CLO01, CLO02, CLO05	Think – Pair- Share, Peer Review	Courseera-Python for Everybody Specialization (Getting Started with Python)
2	Data Types in Python, Loops, Strings, List	CLO01, CLO02, CLO04	Think – Pair- Share, Brain Storming sessions	Courseera- Crash Course on Python

10. Delivery Details of Content Beyond Syllabus

Content beyond syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly.

S.No	Advanced Topics, Additional Reading, Research papers and any	CLO	POs	ALM	References/MOOCs
1	Fundamentals of python library: Numpy : Array creation, Adding ,removing, and sorting elements of array, array dimension, reshaping, 1D, 2D arrays	CLO02, CLO03	PO1, PO2, PO3, PO4, PO5, PO12	Think – Pair- Share, Peer Review	https://numpy.org/doc/stable/user/basics.html https://numpy.org/doc/stable/user/absolute_beginners.html
2.	Fundamentals of python library: Pandas	CLO02, CLO03, CLO04	PO1, PO2, PO3, PO4, PO5, PO12	Think – Pair- Share, Peer Review, Brain storming	https://pandas.pydata.org/docs/user_guide/index.html

11. Evaluation Scheme & Components:

Assessment Type	Evaluation Component	Type of Component	No. of Assessments	% Weightage of Component	Max. Marks	Mode of Assessment	CLO
Formative	Component1	Continuous Lab Evaluations	01*	15%	15	Practical, Viva, File	CLO01-CLO06
Summative	Component2	Sessional Tests(STs)	02**	35%	35	Computer Based Test	CLO01-CLO06
Summative	Component3	End Term Examination	01***	50%	50	Computer Based Test	CLO01-CLO06
Total			100%				

Component	Requirement to Pass
Internal Assessments (Component 1 & Component 2)	Minimum 50% of the total internal marks (STs and other internal evaluations)
End-Term Examination (Component 3)	Minimum 50% of the total end-term marks
Overall Requirement	Must pass both components individually

* Continues Evaluation (CE) is a mandatory evaluation taken once in a semester.

**All 02 STs are mandatory and taken 2 times in a semester.

***As per academic guidelines, a minimum of 75% attendance is required to appear in the end-of-semester examination.

Note: A student securing less than **50%** in either internal or end-term exam will be considered **fail**, even if the combined aggregate is **50% or above**.


12. Syllabus of the Course:

Subject:			
S.No.	Topic(s)	No. of Lectures	Weightage %
1	Introduction to Python programming, Python environment setup, identifiers, reserved words ,lines and indentation, multiline statements, comments, quotation, input/output, output formatting(Integers and string). Fundamentals: Variables(simple, assigning multiple values, output variables, global/local variables), Python decision making- If else and Nested, If else and elif. Match statement Introduction to while and for loop, Concept of nested looping. For loop and range method, Break, Continue and Pass, Pattern Design,	10	30%
2	Defining and calling Functions, Type of arguments (required, keyword, default, variable length), Scope Rules(Global/local Statements), Passing Arguments(call by value/call by reference), Recursive Functions, Lambda Expressions. Lists: Introduction to Lists, Operations on Lists, Storage Structure, Resizing and Looping on Lists, Indexing and Slicing, Passing variables through Functions, passing Lists through Functions, List Comprehension, map, filter & reduce functions, Two Dimensional Lists: Introduction and Storage, Jagged Lists, String- Introduction, methods, comprehension and its relative properties, Tuples, Set and Dictionaries- Introduction, methods, comprehension and its relative properties.	12	30%
3	Modules: Creating Modules, Variables in Modules, Imports and Attributes, Namespaces Reloading, Built in modules, Generating Random values, Files and Directories, File I/O, File positioning, File operators, Exception Handling: Introduction to Exception, Exception Handling, Except clause, Try, finally clause, User Defined Exceptions.	12	25%
4	Exploring Object-Oriented Programming In Python-Introduction, Creating classes, objects, attributes. Constructors., Idea of inheritance, Superclass And SubClass. Overriding methods, Object class , Instancefunction.	10	15%

13. Academic Integrity Policy:

Education at Chitkara University builds on the principle that excellence requires freedom where Honesty and integrity are its prerequisites. Academic honesty in the advancement of knowledge requires that all students and Faculty respect the integrity of one another's work and recognize the importance of acknowledging and safeguarding intellectual property. Any breach of the same will be tantamount to severe academic penalties.

This Document is approved by:

Designation	Name	Signature
Course Coordinator	Dr. Vibhu Sharma	
Program In charge	Dr. Preetinder Singh Brar	
Pro Vice Chancellor	Dr. Jaiteg Singh	
Date (DD/MM/YYYY)	15/07/2025	